Shihao Weng

Ph.D. Student, Nanjing University

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My research is on **trustworthy AI**: making large language models and agents reliable, safe, and aligned with human values. I work across the model lifecycle, from how models are trained and aligned, to how their behavior is evaluated and understood, to how they are safeguarded once deployed in the real world.

Education

Nanjing University 2024 – present
Ph.D. in Software Engineering, advised by Prof. Yang Feng

Nanjing University 2022 – 2024
M.S. in Software Engineering

Jiangnan University 2018 – 2022
B.E. in Computer Science and Technology

Research Experience

ByteDance Nov 2026 – present
Research Intern

The Chinese University of Hong Kong Oct 2026 – present
Visiting Ph.D. Student, hosted by Prof. Michael R. Lyu

Singapore Management University Sep 2025 – Sep 2026
Visiting Ph.D. Student, hosted by Prof. Xiaofei Xie

Publications

† equal contribution * corresponding author

Preprints

P1 **When Code Becomes an LLM Call: An Empirical Study of Call-Site Contracts in Agentic Systems**
Shihao Weng†, Longtian Wang†, Yang Feng*, Mingfei Cheng, Jia Liu, Chao Shen, He Ye, Federica Sarro, Xiaofei Xie
arXiv:2610.24545, 2026 paper

P2 **AgentAntibody: An Adaptive Immune System for Defending LLM Agents against Prompt Injection**
Shihao Weng, Yang Feng*, Xiaofei Xie, Jiongchi Yu
arXiv:2608.04053, 2026 paper

P3 **ARGUS: Defending LLM Agents Against Context-Aware Prompt Injection**
Shihao Weng, Yang Feng*, Jinrui Zhang, Xiaofei Xie, Jiongchi Yu, Jia Liu
arXiv:2605.03378, 2026 paper code

Conference Papers

C1 **Are LLM Safety Judges Policy-Invariant? A Three-Principle Stress-Test**
Shihao Weng, Yang Feng*, Xiaofei Xie
Conference on Neural Information Processing Systems (NeurIPS 2026, E&D Track) **Spotlight** paper code

C2 **TACO: Trust Assessment of Large Language Models in Coding Assistance Tasks**
Shihao Weng, Yang Feng*, Jincheng Li, Yining Yin, Zhenlun Zhang, Lyuxi Liu, Jia Liu
IEEE/ACM International Conference on Software Engineering (ICSE 2026) paper code tool

C3 **AtPatch: Debugging Transformers via Hot-Fixing Over-Attention**

Shihao Weng, Yang Feng*, Jincheng Li, Yining Yin, Xiaofei Xie, Jia Liu

IEEE/ACM International Conference on Software Engineering (ICSE 2026) paper code

C4 **VADA: A Multicultural Benchmark for Value-Aware Data Generation and Alignment Evaluation in LLMs**

Zhenlun Zhang, Yang Feng*, **Shihao Weng**, Yining Yin, Jincheng Li, Jia Liu

IEEE/ACM International Conference on Software Engineering (ICSE 2026) paper code

C5 **Lightweight Probabilistic Coverage Metrics for Efficient Testing of Deep Neural Networks**

Yining Yin, Yang Feng*, **Shihao Weng**, Xinyu Gao, Jia Liu, Zhihong Zhao

Asia-Pacific Symposium on Internetware (Internetware 2025) paper code

C6 **Datactive: Data Fault Localization for Object Detection Systems**

Yining Yin, Yang Feng*, **Shihao Weng**, Yuan Yao, Jia Liu, Zhihong Zhao

ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2024) paper code

C7 **Dynamic Data Fault Localization for Deep Neural Networks**

Yining Yin, Yang Feng*, **Shihao Weng**, Zixi Liu, Yuan Yao, Yichi Zhang, Zhihong Zhao, Zhenyu Chen

ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE 2023) paper code

C8 **Prioritizing Testing Instances to Enhance the Robustness of Object Detection Systems**

Shihao Weng, Yang Feng*, Yining Yin, Jia Liu

Asia-Pacific Symposium on Internetware (Internetware 2023) **Distinguished Paper Award** paper code

Journal Articles

J1 **Data Preparation and Quality for Code-Centric Generative Software Engineering Tasks: A Systematic Literature Review**

Shihao Weng, Yang Feng*, Yining Yin, Zhenlun Zhang, Baowen Xu

Frontiers of Computer Science, 2025 paper

J2 **Seeing the Invisible: Test Prioritization for Object Detection System**

Shihao Weng, Yang Feng*, Yining Yin, Yuxuan Dai, Jia Liu, Zhihong Zhao

Empirical Software Engineering 29(6):154, 2024 paper code

Honors and Awards

Outstanding Graduate Student, Nanjing University (two consecutive years) 2023 – 2024

Distinguished Paper Award, Internetware 2023 2023

National Scholarship for Graduate Students, Ministry of Education of China 2023

CCF Outstanding Undergraduate Student, China Computer Federation 2021

Bronze Medal, 45th ICPC Asia Regional Contest 2021

National Scholarship for Undergraduate Students (two consecutive years) 2020 – 2021

Academic Service

Reviewer, NeurIPS 2026 Workshop (TAE) 2026